Course: SWE-Computer Network Practical							
Instructor	Dr. Asad Raza Malik		Practical/Lab No.		12		
Date			CLOs		03		
Student's Roll no:			Point Scored:				
Date of Conduct:			Teacher's Signature:				
LAB PERFORMANCE INDICATOR	Subject Knowledge	Data Analysis and Interpretation	Ability to Conduct Experiment	Presentation	Calculation and Coding	Observation/ Result	Score

Topic	To configure the IGRP dynamic routing protocol on two routers
Objective	• The objective of this lab is to configure the IGRP dynamic routing protocol on two routers to enable communication between different network segments. • Students will learn how to set up and verify IGRP, understand its behavior in sharing routing information, and analyze its advantages and limitations compared to other routing protocols. • This lab aims to develop skills in implementing distance-vector routing protocols and managing larger-scale network environments.

Lab Discussion: Theoretical concepts and Procedural steps

Theoretical Background:

What is IGRP?

- IGRP (Interior Gateway Routing Protocol) is a distance-vector routing protocol developed by Cisco to overcome the limitations of RIP (Routing Information Protocol). It provides better scalability and more efficient handling of routing in medium-sized and large networks.
- IGRP supports multiple metrics to determine the best path, including bandwidth, delay, load, and reliability, making it more accurate than RIP.

Key Characteristics of IGRP:

- **Classful Protocol:** IGRP is classful, which means it does not support subnet masks in its routing updates and does not support Variable Length Subnet Masking (VLSM).
- **Metrics Used:** It uses a combination of metrics like bandwidth, delay, load, and reliability to calculate the best path to a destination, unlike RIP, which uses only hop count.

- **Administrative Distance:** IGRP has an administrative distance of 100, making it more trustworthy than RIP but less than EIGRP (Enhanced Interior Gateway Routing Protocol).
- **Maximum Hop Count:** IGRP supports a maximum hop count of 255, compared to RIP's 15-hop limit.

Timers Used in IGRP:

- **Update Timer:** IGRP sends its routing updates every 90 seconds.
- **Invalid Timer:** A route is considered invalid if no updates are received for 270 seconds.
- **Hold-down Timer:** Prevents route changes for 280 seconds to stabilize the routing environment.
- **Flush Timer:** Routes are removed from the table after 630 seconds if no updates are received.

Functions

The IGRP performs a variety of functions:

Interior Gateway Routing Protocol (IGRP) was developed by Cisco in response to the restrictions of the Routing Information Protocol (RIP), which manages a maximum hop count of 15 per connection. The Interior Gateway Routing Protocol (IGRP) allows for a maximum hop count of 255. The fundamental two objectives of the IGRP are as follows:

1. Route information should be sent between all linked routers inside its border or autonomous system.
2. Continue to update anytime there is a topological, network, or route change that takes place.

Every 90 seconds, the IGRP broadcasts to its neighbors a notice of any new modifications as well as information about its current condition.

IGRP is responsible for maintaining a routing table with the most optimum route to the corresponding nodes and networks inside the parent network, as determined by the parent network. Given that it is a distance-vector protocol, the IGRP calculates the metric for the shortest route to a certain destination based on a number of different criteria.

Advantages:

1. The procedure is simple and uncomplicated.
2. It considers the latency, bandwidth, reliability, and load of a network connection while calculating the score. Consequently, it is quite accurate when it comes to selecting the most suited approach.
3. The use of composite metrics
4. Configuration is straightforward.
5. When compared to RIP, it has more scalability (255 hops, 100 by default)

Disadvantages:

1. The hop count is limited to 15; if a packet has traveled through 15 routers and still has another router to travel to, it will be discarded.
2. Does not support a variable-length subnet mask (VLSM), which means that it sends routing updates based only on a fixed-length subnet mask (FLSM) or routes that fall on classful boundaries. As a result, RIP V1 will not function on a network that has been subnetted beyond the standard /8, /16, and /24 (255.0.0.0, 255.255.255.0) or Class A, B, and C network borders (255.0.0.0, 255.255.255.0).
3. Convergence occurs slowly, particularly on large networks.
4. Doesn't know how much bandwidth is available on a given connection.
5. Doesn't allow numerous pathways for the same route
6. Routing updates may use a substantial amount of bandwidth since the whole routing database is delivered whenever the state of a connection changes. Routers are prone to routing loops.

Lab Steps: Configuring IGRP on Two Routers

Step 1: Network Setup

1. Set up a network topology with two routers, Router A and Router B.
2. Connect Router A to one network segment (e.g., 192.168.10.0/24) and Router B to another network segment (e.g., 192.168.20.0/24).
3. Use a common network link between the two routers, such as the 172.16.0.0/16 network.

Step 2: Assign IP Addresses to Interfaces

Assign IP addresses to all the router interfaces in the topology. Make sure the IP addressing scheme is consistent with the network segments.

Step 3: Configure IGRP on Both Routers

Router A Configuration Example:

```
Router> enable
Router# configure terminal
Router(config)# router igrp 10
Router(config-router)# network 192.168.10.0
Router(config-router)# network 172.16.0.0
Router(config-router)# exit
Router(config)# exit
Router# write memory
```

Router B Configuration Example:

```
Router> enable
Router# configure terminal
Router(config)# router igrp 10
Router(config-router)# network 192.168.20.0
Router(config-router)# network 172.16.0.0
Router(config-router)# exit
Router(config)# exit
Router# write memory
```

Explanation of the Commands:

- router igrp 10: Starts the IGRP routing process, where 10 is the Autonomous System (AS) number.
- network [network address]: Specifies the directly connected networks that IGRP should advertise.
- write memory: Saves the configuration to the router's NVRAM.

Verification of IGRP Configuration

1. **Check the Routing Table:**
    - Use the command show ip route to view the routing table on both routers. Verify that routes learned via IGRP are present.
2. **Verify IGRP Settings:**
    - Use show ip protocols to check the protocol settings, including timers and routing updates.
3. **Debugging IGRP:**
    - Use debug ip igrp transactions to monitor real-time IGRP routing updates.
    - Observe how IGRP converges when a link goes down and how it chooses the new path based on metrics like delay and bandwidth.

Analysis and Discussion

- **Analyze the Benefits of IGRP:** Compare IGRP's multiple metric-based routing with RIP's single metric (hop count) to understand its efficiency.
- **Scalability**: Discuss how the 255-hop limit of IGRP enables it to handle larger networks compared to RIP.
- **Limitations of IGRP**: Explain how the lack of support for VLSM can limit its deployment in modern networks.

Assignment Questions for Students

1. What are the primary differences between IGRP and RIP?
2. Why is IGRP considered more suitable for larger networks compared to RIP?
3. Explain how the different metrics used by IGRP improve the routing decision process.
4. What would happen to IGRP updates if there is a failure in the network? How does the protocol handle convergence?

Submission Requirements:

- **Configuration Screenshots**: Provide screenshots of the IGRP configuration on both routers.
- **Routing Table Output**: Include a snapshot of the routing tables before and after the link failure test.
- **Analysis Report**: Write a 1-2 page report summarizing the setup, verification steps, and the behavior of IGRP during convergence.
- **Answer the Assignment Questions** in a separate document.

Grading Criteria:

- **Configuration (40 points)**: Correct configuration of IGRP on both routers.
- **Verification (20 points)**: Proper verification and debugging of the routing tables.
- **Analysis (30 points)**: Thorough understanding of IGRP's behavior, its advantages, and limitations.
- **Report Quality (10 points)**: Clarity, organization, and completeness of the report.

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